LOCO WATERY COMPANY

SINCE 2025

ANALYSIS REPORT

INTRODUCTION TO ENVIRONMENTAL MODELLING

WETLAND'S WATER ABSTRACTION MODEL

EXAM NO. B265609

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INTRODUCTION

1.1 Context and Background

Loco Watery, a local water company in Scotland, sought to abstract some water from the river that feed the Loco Wetland due to a shortage supply in the surrounding area. Abstraction provides essential water for public water supply, agriculture and industry. However, unsustainable levels of abstraction impact the ecology and resilience of our rivers, wetlands and aquifers (GOV.UK, 2017). Therefore, the company must submit a proposal to the local government along with their impact assessment analysis on the wetland.

Wetlands are an important part of the ecosystem performing as nature-based solutions (NBS) for water management (Ferreira *et al.*, 2023). They can enhance resilience of a catchment to floods and torrents and safeguard water availability during droughts, remove or reduce a number of organic and inorganic pollutants (e.g., nutrients, heavy metals, hydrocarbons, pesticides), and they have proven to be efficient and effective in improving the quality of water from different sources, such as runoff from agriculture and urban areas, and domestic and industrial wastewater. (Ferreira *et al.*, 2023)

Having the right flow in our rivers and protecting groundwater levels is essential to supporting healthy ecology, enhancing natural resilience to drought. Sustainable water abstraction is therefore essential to ensure that river flows and groundwater levels support ecology and natural resilience (GOV.UK, 2017).

1.2 Objective

The primary aim of this simulation is to determine the maximum sustainable volume of water that can be extracted from the wetland while maintaining its ecological balance and prevent a serious distress to its biomass conservation value.

MODEL DESCRIPTION

2.1 Software Used

For this purpose, we are using OpenModel software. This software is designed for practical and simple visualisation, parameterisation, and numerical analysis. It's Ideal for dynamic models that use ordinary differential equations (ODEs) and simple assignments through their modular structure easy equation scripting, and outputs in tabular and graphical formats (Crout, 2020).

While other software alternatives, such as Simile and HEC-RAS, can be used for modeling similar scenario, the simplicity and ease of checking syntax errors make OpenModel more straightforward. Simile, although a powerful tool with a visual interface and strong capabilities for ecological modeling (Muetzelfeldt and Massheder, 2003), would likely require a longer time to evaluate and run the model. Furthermore, HEC-RAS is excellent for simulating water flow and analyzing water quality in river systems (Zainal and Abu Talib, 2024), however it has limited capabilities for modeling the ecological processes essential to a wetland ecosystem, such as biomass growth making it less appropriate for the objective of this specific wetland model compared to the focus on simplicity and ease of use offered by OpenModel.

2.2 Scope of the Model

The model would estimate changes in wetland biomass through calculations of biomass growth and loss. The growth or loss of biomass influenced by water and vegetation factors such as temperature, solar radiation, water input from the river inflow and other things detailed in **Table 1**. The model is run for 3287 timesteps, which is adjusting to the daily weather and water inflow data we have as data source. We also have received data reference for the relationship between water content (θ) with outflow rate and evaporation.

2.3 Assumptions and simplifications

Modeling realistic wetland conditions is challenging due to variable surface water visibility and diverse vegetation types. Including key attributes like (i) permanent/seasonal water presence, (ii) waterlogged soils with organic accumulation, and (iii) water-adapted flora (and fauna) further increases complexity (McManamay *et al.*, 2018).

Thus, we've made several assumptions to create a simple model that represents a wetland biomass fluctuation. First, we assume water input comes from just one river, which also serves as the source for water abstraction. For vegetation, we use a single generic plant 'type' to drive calculations for leaf area index, fraction interception, growth rates, and biomass changes. The modelled wetland covers 10 km² with 1 m soil depth, giving a saturated water volume of 10 million m³. To reach our objective, we've set a conservation threshold where biomass dropping below 50% of normal levels is considered critical. Initial conditions include 85 g/m² biomass, theta of 1, and full water storage at 10,000,000 m³.

2.4 Limitations of the model

This model has several important constraints. First, it only uses historical biomass data and doesn't include future projections, creating uncertainty for planning water abstraction. Second, the timesteps in the given datasets lack actual dates, making it impossible to align water extraction strategies with specific seasons. Third, the model simplifies reality by ignoring key ecological factors like land use changes, human disturbances, soil properties, and species diversity. Due to these simplifications, the model is best suited for basic simulations of Loco Wetland's biomass dynamics rather than precise functional model and is even less suitable for use in other wetland models.

2.5 Conceptual Model

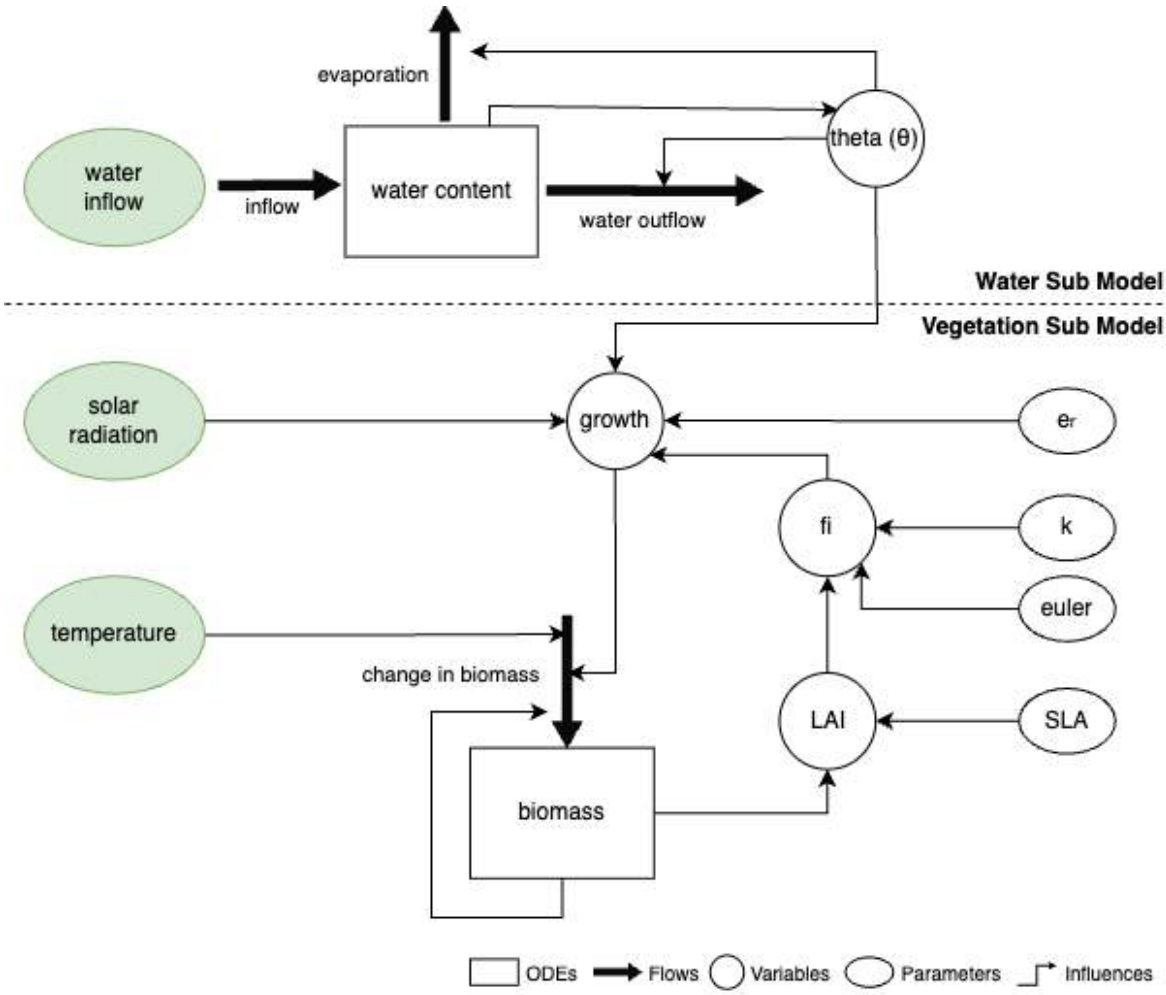

***Figure 1**. Loco Wetland Biomass and Water Content changes model*

To understand the model better, we created a conceptual diagram (**Figure 1**) showing the key relationships between components. The model simulates wetland biomass growth, where biomass changes are influenced by temperature, growth factor and the current biomass level itself. This growth factor is calculated using solar radiation, a conversion coefficient, fraction interception, and theta (the ratio of water volume to soil volume, in m^3/m^3). Theta, a volumetric water content is determined by water content, which changes based on river inflows minus losses from evaporation and outflow.

Importantly, theta directly affects both evaporation and outflow rates. We've divided these calculations into two interconnected sub-models: water content (detailed in Section 2.5) and vegetation (Section 2.6). Some parameters remain constant, while others (like the green-highlighted inputs in **Figure 1** - including river inflow data, daily solar radiation, and temperature) update with each timestep based on the given data sources. These dynamic inputs will drive the model's calculations in subsequent steps.

Table 1. Factors Used in the Loco Wetland Model

Category	Name	Symbol	Description
ODEs	Biomass	biomass	Total mass of vegetations in a given area (g/m²)
	Water Content	WC	Volume of water present in the wetland (m³)
Variables	Growth	growth	The rate at which vegetation biomass increases
	Leaf Area Index	LAI	Total leaf area coverage per unit of ground area (m² leaf/m² ground)
	Fraction Interception	fi	The proportion of solar radiation that is intercepted by the plant canopy
	Theta (θ)	theta	The amount of water per unit amount of soil (m³ of water / m³ of soil),
	Water outflow	outflow	The rate at which water leaves the wetland (m³)
	Evaporation	evaporation	Amount of water loss from the wetland surface (mm)
Parameters	Temperature	temp	Atmospheric temperature (°C)
	Solar Radiation	S	The amount of daily solar energy received per unit area (MJ/m²)
	Conversion Coefficient	e_r	A constant value representing the efficiency of converting intercepted solar radiation into plant biomass (1.4 g/MJ)
	Specific Leaf Area	SLA	A vegetation-specific characteristic indicating the leaf area per unit of leaf mass (0.02 m² / g)
	Extinction coefficient	k	The ratio of light (solar radiation) decreases as it passes through the layers of leaves in the plant canopy (0.45)
	Euler number	euler	A mathematical constant used in the equation for fraction interception. (2.72)
	Water inflow	inflow	The amount of water enters the wetland from the river (m³/d)

2.6 Vegetation Sub-Model

The model assesses conservation value through daily biomass changes, calculated using **Equation 1**. All factors used here are defined in **Table 1.**

$$\frac{\Delta biomass}{\Delta t} = growth - \frac{temp}{300} \times biomass \quad (1)$$

Biomass changes results from growth (**Equation 2**) and senescence — the temperature-dependent loss of plant material from dying leaves and other parts. For senescence, we calculate daily losses by multiplying the previous timestep's biomass by temp/300 (d^{-1}), where this coefficient accounts for our generic plant type. Daily temperature data comes from given datasets.

$$growth = fi \times e_r \times S \times \theta \quad (2)$$

The growth rate formula is influenced by three key variables such as the fraction of solar radiation intercepted represented *fi* (**Equation 3**). Then, there is e_r, which is a constant value of solar radiation conversion efficiency multiplied by the daily solar radiation amount from the data source given and the theta (θ) that will be explained further in the water sub model (**Equation 5**).

$$fi = 1 - e^{-k \times LAI} \quad (3)$$

For *fi* calculation, we need Euler's number (2.72), extinction coefficient (0.45), and the canopy leaf area index (LAI). LAI can be obtained by using **Equation 4.**

$$LAI = 0.5 \times biomass \times SLA \quad (4)$$

We use 0.5 because we assume leaves make up half of the total plant material. SLA simply means how much leaf area you get per gram of leaves.

2.7 Water Sub-Model

The water sub-model affects vegetation growth through theta (θ), which represents water saturation in the soil. Theta is calculated using **Equation 5**:

$$theta = \frac{WC}{10{,}000{,}000} \quad (5)$$

Here, 10 million m³ comes from multiplying our wetland area (10 km²) by soil depth (1 m). This means:

- When WC ≤ 10 million m³: θ equals the actual water proportion
- When WC > 10 million m³: θ exceeds 1, but its effect on vegetation growth caps at 1

We implement this in **Syntax 1** (line 9) with an "if" statement to maintain the growth factor at 1 even when θ > 1, as observed in our data.

In relation to our theta calculations, we model water content (WC) changes using **Equation 6**:

$$\frac{\Delta WC}{\Delta t} = inflow - outflow - evaporation \quad (6)$$

To determine evaporation and outflow rates, we analyzed two relationships, theta against evaporation and theta against outflow rate. Using linear regression on these relationships (shown in the Appendix), we derived predictive equations for both processes. A key consideration was the evaporation data's millimeter units — to convert this to water loss volume for our wetland model, we multiply the evaporation depth (in meters) by the wetland's 10 km² area. This means 1 mm of evaporation equals 10,000 m³ of water loss (0.001 m × 10,000,000 m²).

With all units standardized, the calculation will be as shown in **Equation 7**:

$$\frac{\Delta WC}{\Delta t} = inflow - (4818.5 \times theta - 2354.5) - (9598.1 \times theta - 1221.5) \quad (7)$$

Because we're using OpenModel, we need to implement the differential equations for biomass and water content using the '.rate' suffix. Another important OpenModel command to use is when we're going extract inflow, solar radiation, and temperature data from the provided spreadsheets. We use the format weather.s(t), which reads values from column S in the 'weather' sheet for each time step t. The complete syntax for our OpenModel 'Main' window appears in **Syntax 1** below.

Syntax 1. Complete Wetland Model

```
1  // Biomass
2    // Vegetation sub model
3      // Determine the change in biomass
4      biomass.rate = growth - (weather.temp(t)/300) * biomass
5
6      // Growth rate of vegetation biomass (g/m^2) restricted by reduction in wetland
          water content
7      growth = fi * er * (weather.S(t)) * theta
8
9      if theta > 1
10     theta = 1
11     endif
12
13     // fraction of interception
14     fi = 1 - euler ^(-k*LAI)
15
16     // Leaf Area Index
17     LAI = 0.5 * biomass * SLA
18
19   // Water sub model
20     // Rate of change in Water Content (Water Inflow - Outflow - Evaporation)
21     WC.rate = (WaterData.Inf(t) * (1 - water abstraction rate)) - (4818.5 * theta -
       2354.5) - (9598.1 * theta - 1221.5)
22
23     // theta calculation (m^3 of water / m^3 of soil)
24     theta = WC / 10000000
```

Using the established syntax, we can run the model to analyze suitable water abstraction rates. The process involves modifying the 'water abstraction rate' value in **Syntax 1 (line 21)** for each scenario we want to test. We began with the baseline condition (water abstraction rate = 0) and progressively increased the extraction up to 50% (rate = 0.5). Between these extremes, we tested 10 different scenarios at regular 5% intervals (0%, 5%, 10%, ..., 50%) to systematically evaluate how each extraction level impacts the wetland system.

MODEL VALIDATION

We validated our normal model outputs against the observed biomass data provided in the dataset. The validation covers timesteps 400 through 1120, with biomass measurements available at 20-day intervals.

3.1 One-to-one graph validation

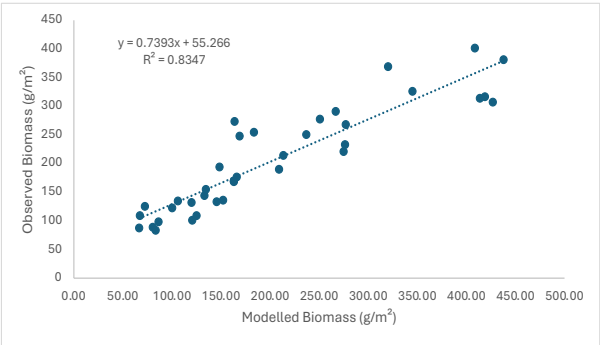

***Figure 2**. Comparison of Modelled vs. Observed Biomass Values*

The model shows good alignment with the observed data, as can be seen in R-squared value of 83.5%. This means it explains most of the variation we see in the actual observation. When we plot the model results against the real data points, they line up nicely. To check this further, we'll add statistical tests and look at time series graphs.

3.2 Statistical Method Validation

Table 2. Model Validation Statistics: RMSE, CV(RMSE), and R² Values

Statistic	Value
RMSE	46.81333
CV(RMSE)	23%
R - Squared	0.82345

Using statistical methods, the validation results also show that the model performs well, with an R-squared value of 0.82345, supporting the one-to-one graph validation with similar value. This means the model explains approximately 82% of the variation in the observed biomass. The root mean square error (RMSE) was 46.81, indicating the average difference between predicted and actual values. The CV(RMSE) of 23% means that, on average, the model's prediction error is about 23% of the typical observed biomass value, which shows the errors are relatively small compared to the actual data.

3.3 Time Series Graph Validation

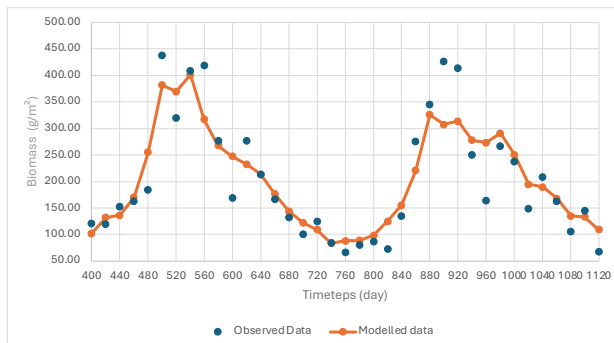

***Figure 3**. Modelled Biomass and Observed Biomass Over Time*

The time-series graph shows how the modelled and observed biomass values change over time. Overall, the model follows the general trend of the observed data quite well. We can also see that the model predictions tend to match the observed values more closely when the biomass is lower, suggesting the model performs best during periods of lower vegetation growth. While some differences appear when biomass is higher, the time-series comparison still shows good agreement, supporting the model's ability to capture biomass dynamics over time.

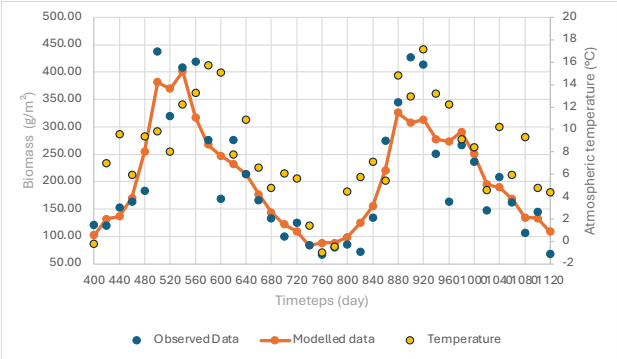

***Figure 4**. Modelled Biomass and Observed Biomass Over Time Compared with Temperature*

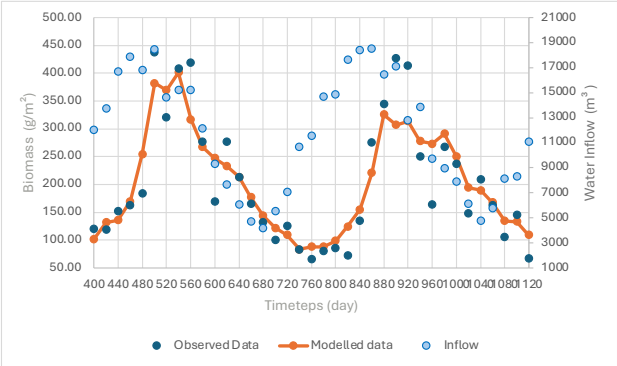

***Figure 5**. Modelled Biomass and Observed Biomass Over Time Compared with Water Inflow*

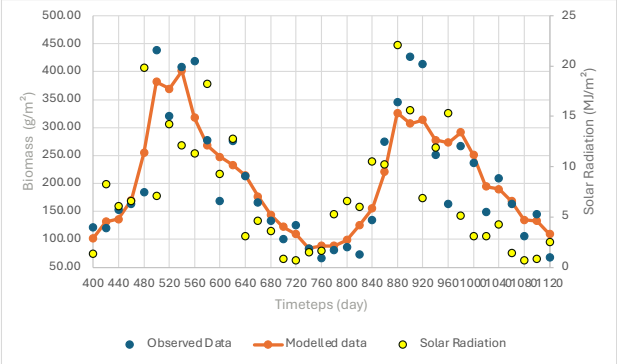

***Figure 6**. Modelled Biomass and Observed Biomass Over Time Compared with Solar Radiation*

Additionally, **Figures 4, 5, and 6** show the modelled and observed biomass over time compared with three key parameters. temperature, water inflow, and solar radiation, each plotted on a secondary Y-axis. These graphs help us visually explore how well the model responds under different conditions. In all three figures, we notice that the model matches the observed biomass more closely when the influencing factors are lower. This suggests the model performs better under less extreme or less variable environmental conditions. However, since we didn't conduct a full sensitivity analysis, we can't say for sure which factor has the strongest influence on model accuracy.

RESULTS

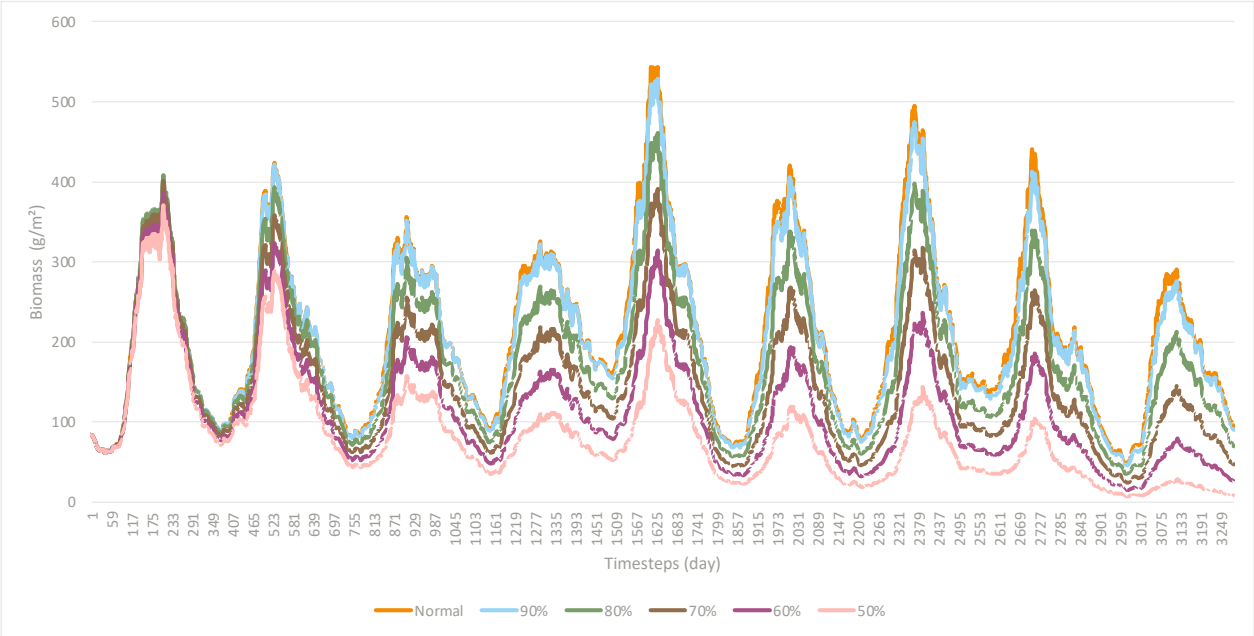

Figure 7. *Wetland Biomass Balance Comparison: With vs. Without Water Abstraction*

After running multiple model simulations, we compiled all results into a single comparative chart **Figure 5** for clearer visualisation of the different water abstraction scenarios. This combined plot allows us to directly observe how varying extraction rates impact wetland biomass over time.

To determine the optimal water abstraction, we developed three extraction schemes: short-term (3 years), mid-term (6 years), and long-term (9 years) to evaluate sustainable water withdrawal. Using a 9-year simulation (total timesteps ÷ 365 days/year), we set assessment points at the end of every three year, timesteps 1096 (short-term), 2192 (mid-term), and 3287 (long-term). These scenarios help Loco Water Company determine safe extraction limits before exceeding the 50% biomass loss threshold, with selection depending on water demand urgency

Table 3 directly compares the maximum extraction rates for each scheme, alongside remaining biomass levels:

- Short-term allows up to 40% abstraction
- Mid-term allows up to 35% abstraction
- Long-term allows a maximum of 25% abstraction

Table 3. Percentage Biomass Loss Under Different Water Extraction Scenarios

Water Extraction	End of Year 3		End of Year 6		End of Year 9	
	Biomass (g/m²)	% Loss	Biomass (g/m²)	%Loss	Biomass (g/m²)	%Loss
Normal	134.86	0.0%	103.29	0.0%	94.067	0.0%
5%	134.86	0.0%	103.29	0.0%	94.067	0.0%
10%	133.35	-1.1%	99.311	-3.9%	89.404	-5.0%
15%	125.19	-7.2%	89.290	-13.6%	78.932	-16.1%
20%	115.36	-14.5%	79.793	-22.7%	68.395	-27.3%
25%	105.59	-21.7%	70.413	-31.8%	57.675	-38.7%
30%	95.951	-28.9%	61.164	-40.8%	46.783	-50.3%
35%	86.295	-36.0%	52.026	-49.6%	35.830	-61.9%
40%	76.599	-43.2%	43.004	-58.4%	25.173	-73.2%
45%	66.894	-50.4%	34.135	-67.0%	15.546	-83.5%
50%	57.238	-57.6%	25.521	-75.3%	7.9581	-91.5%

Red highlighted cells in the table indicate when biomass depletion exceeds the critical 50% threshold, signaling unsustainable extraction levels. These visual alerts help quickly identify when abstraction rates would compromise the wetland's conservation value.

The remaining biomass levels shown for each scenario demonstrate the cumulative impact of different extraction rates over time. This comparative analysis enables Loco Water Company to balance water supply needs with ecological preservation.

The safest option is taking just 5% of water, since this doesn't reduce the biomass at all, even after nine years of daily extraction. While this gives Loco Water Company a safe starting point, they'll need to consider these biomass levels along with other factors when making final decisions. More detailed analysis is still needed to confirm these findings and determine the best water extraction plan that balances supply needs with the wetland's ecosystem sustainability.

DISCUSSION

Wetland biomass growth follows strong seasonal patterns, peaking in spring and summer due to warmer temperatures, longer daylight, and higher nutrient availability (Choi *et al.*, 2012). During these high-growth periods (May–August) or in more temperate climate (Spring and Summer), increased water extraction may be sustainable, while reduced abstraction should be considered in autumn and winter when growth slows or stops (Gorme *et al.*, 2012). However, our model's lack of precise seasonal dates limits exact timing for these adjustments. Despite this, the results confirm that aligning extraction with natural biomass cycles, rather than using a fixed rate, can better balance water needs with wetland conservation. Future improvements should integrate exact seasonal data to refine these water management strategies.

CONCLUSIONS

Analysis for Loco Watery Company using OpenModel software indicates that sustainable water extraction from Loco Wetland is feasible under several scenarios. If the company only requires water for the short term, a 40% extraction rate remains safe. However, for mid-term and long-term scenarios, lower rates of 35% and 25%, respectively, are necessary to prevent biomass depletion beyond the critical 50% threshold. Notably, a 5% extraction rate poses no ecological risk, even over a nine-year period.

To improve the model, future studies should integrate a more comprehensive set of factors, including the wetland's existing vegetation species, ecological interactions from wildlife and human activity, and additional water inputs such as rainfall beyond the primary river source. Precise dataset timestamps would help better assess seasonal effects, while incorporating soil properties could further refine the model's accuracy. Additionally, forecasting future water inflow and weather patterns would strengthen long-term projections. Finally, examining socio-anthropological dynamics, particularly on how human interacts with the wetland in different seasons, could provide valuable context for sustainable management.

Even better, sensitivity analysis can be implemented for more advanced models, further enhancing their usefulness. By better understanding which variables and parameters have the greatest impact on biomass changes over time, Loco Watery Company can adjust and optimize water abstraction strategies with better precision.

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P.S. AI tools were used to assist with grammar correction and image generation.

APPENDIX

A1. Regression Plots for Evaporation and Outflow Estimation

To estimate evaporation and outflow rates as functions of theta (θ), we performed linear regression using the provided dataset. The plots below show the regression lines fitted to the data points:

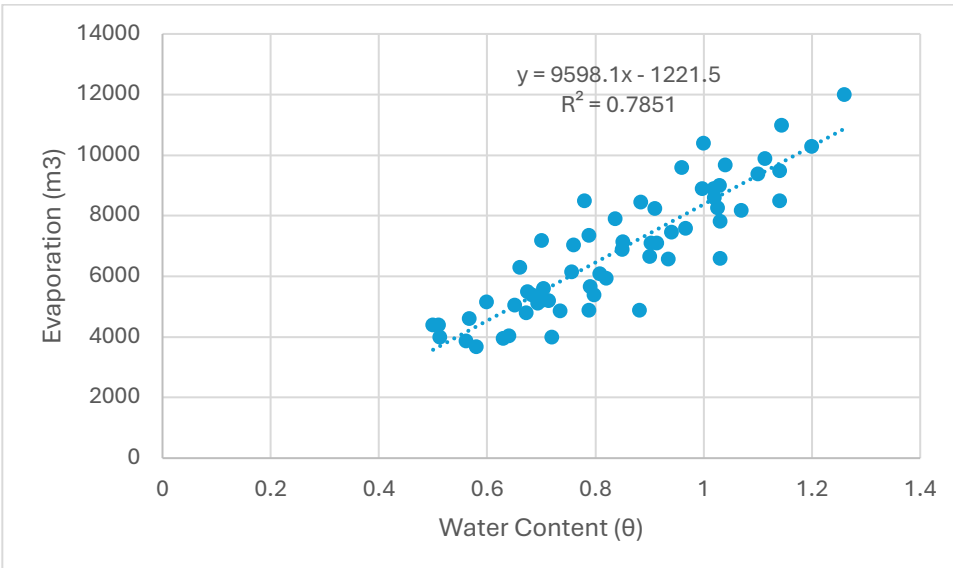

Figure A1. Linear Regression of Theta vs. Evaporation

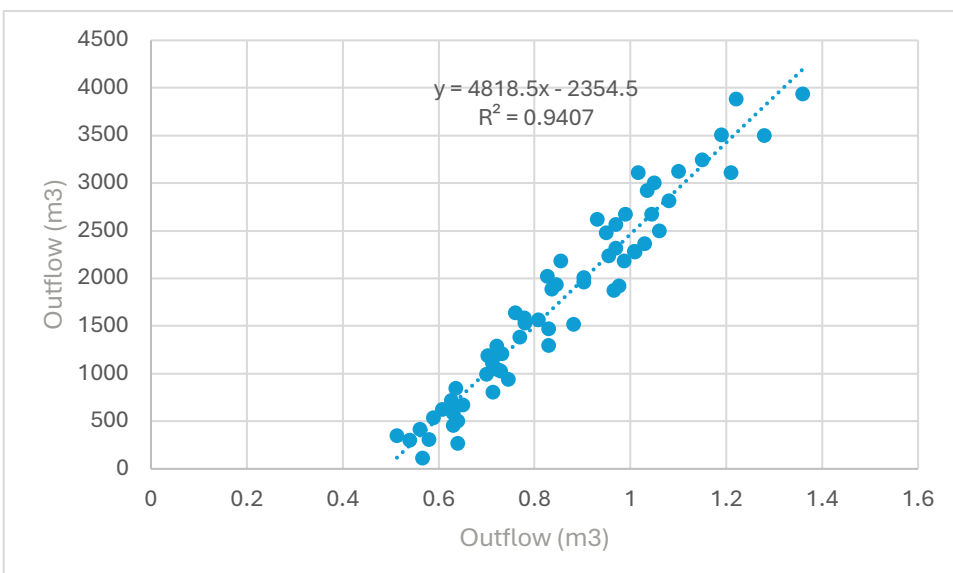

Figure A2. Linear Regression of Theta vs. Outflow Rate

These equations were then used in the model to calculate daily evaporation and outflow volumes as part of the water balance sub-model.